TEAM GRADIENT_ZERO

UNIVERSITY PROJECT · ESP32 + WI-FI CSI

GDZero

Wi-Fi Motion Detection Through Walls

No camera. No sensors. Just the Wi-Fi signals that already fill your room.

Presented by Team Gradient_Zero · github.com/Darwish-V12/GDZero

THE IDEA

Why detect motion without a camera?

Cameras invade privacy

People avoid monitored rooms, and footage becomes a liability.

01

PIR sensors are blind

They need a clear line of sight and only trigger on direct heat, not whole-room presence.

02

Wi-Fi is already everywhere

Router signals bounce through every corner of the home, walls included, all day long.

03

GD Zero listens to how movement bends the Wi-Fi around you.

HOW IT WORKS

Three steps, zero extra hardware

1

Listen

The ESP32 reads the Wi-Fi signal pattern (CSI) coming from your router.

2

Detect

A person moving changes that pattern — the chip notices the disturbance.

3

Show

The live dashboard turns it into a motion alert on a radar view.

ⓘ CSI = Channel State Information — how the signal travels, not what it carries. GD Zero never reads your data.

THE HARDWARE

Everything costs less than a coffee

1× ESP32

the only piece of hardware

about 10$

1× USB cable

data cable, just for setup and power

Any Wi-Fi router

2.4 GHz — one you already own

0 extra sensors

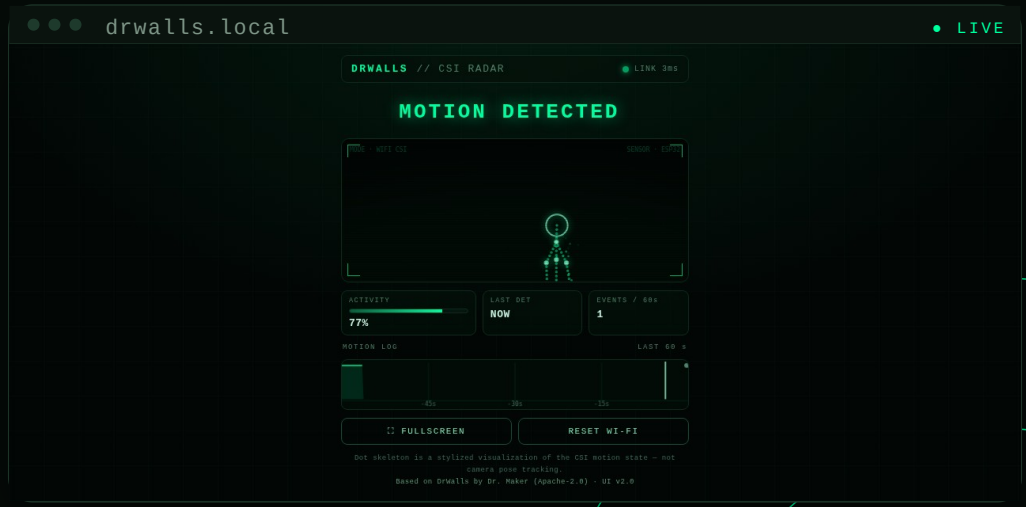
no camera, no PIR, no wiring

If your room already has Wi-Fi, it already has the sensor.

THE INTERFACE

A military-radar dashboard, built right into the device

- A dot-skeleton figure walks the radar while motion is detected
- A live 60-second motion log sits under the radar
- Green LINK badge means the dashboard is connected to the device



MOTION DETECTED — live radar view from the actual device

WHAT MAKES IT FUN

Small device, big dashboard

Dot-skeleton radar

walks on motion, breathes when someone is present.

60s motion log

every detection is stamped on a live timeline.

Auto demo mode

no device reachable? it simulates a target — perfect for practice.

Fullscreen mode

one click turns the whole screen into the radar for presentations.

Runs in the browser

the dashboard is served by the ESP32 itself: no app, no cloud.

Private by design

what you see is a visualization, not a video.

PRIVACY FIRST

A camera sees. GDZero only feels.

Camera

- ✕ Records faces and bodies
- ✕ Stores video somewhere
- ✕ Feels invasive in a home

GDZero

- ✓ Knows only: `someone · moving · quiet`
- ✓ Stores nothing — state lives in RAM
- ✓ No image of you ever exists

"The device never sees you — it only feels the signal change."

WHERE IT FITS

One sensor, many quiet jobs

01 SENSOR · 04 QUIET JOBS

Smart home 01

lights and AC follow the room you are actually in.

Security 02

get alerted when someone enters after hours.

Elderly care 03

daily-activity awareness without a camera in the room.

Energy saving 04

know which rooms are really used, and stop conditioning empty ones.

TRY IT YOURSELF

From zero to radar in 5 minutes

1. **Flash from the browser**

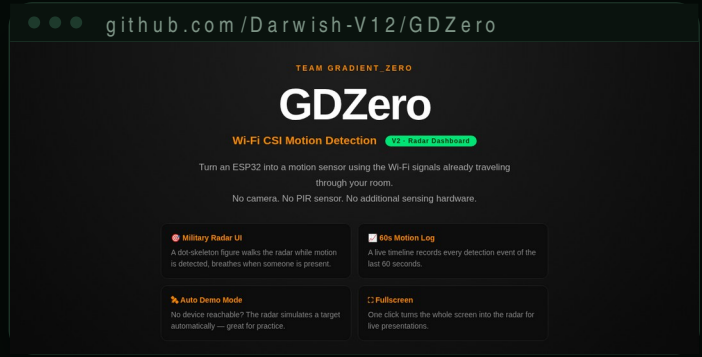
   open the project page in Chrome, connect the ESP32 by USB, click Install. No toolchain, no IDE.

2. **Connect Wi-Fi**

   first boot opens the DrWalls-Setup portal; pick your router once.

3. **Open the radar**

   any browser on the network: `drwalls.local`.



- ONE-CLICK BROWSER INSTALLER — FIRMWARE V2 INCLUDED

⌘ github.com/Darwish-V12/GDZero

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Thank you

Questions? — happy to demo it live.

github.com/Darwish-V12/GDZero

Live dashboard: drwalls.local